

Education

Harvard University

PhD in Statistics; advisor: Marinka Zitnik

Cambridge, MA

2023–Present

University of Oxford

MSc in Statistics; advisor: Yee Whye Teh

Oxford, UK

2021–2023

Stanford University

B.S. in Computer Science; B.A. in Sociology with Honors

Stanford, CA

2017–2021

Publications

* Co-first authors; equal contribution.

Published and accepted

World models for biomedicine. Ayush Noori*, **Nic Fishman***, Ada Fang, Lukas Fessler, and Marinka Zitnik. *Cell* 189 (19), 5845–5869, Sep 17, 2026.

Count Bridges enable Modeling and Deconvolving Transcriptomic Data. **Nic Fishman**, Gokul Gowri, Tanush Kumar, Jiaqi Lu, Valentin De Bortoli, Jonathan S. Gootenberg, and Omar Abudayyeh. *International Conference on Learning Representations*, Apr 2026.

Rethinking Repeatability in Observational Social Science. Jonathan Ben-Menachem, Ari Galper, and **Nic Fishman**. *Sociological Methods & Research (accepted)*, 2026.

The Virtual Embryo Challenge: Generative Modeling of Embryogenesis Across Space, Scale and Time. Jiayuan Ding, Yifan Lu, Qingquan Zhang, Zehua Zeng, Weize Xu, **Nic Fishman**, You He, Siyu He, Le Song, Eric Xing, James Zou, Emily Fox, Marinka Zitnik, Neil Chi, and Xiaojie Qiu. *NeurIPS 2026 Competition Track (accepted)*, 2026.

Generative Distribution Embeddings. **Nic Fishman***, Gokul Gowri*, Peng Yin, Jonathan Gootenberg, and Omar Abudayyeh. *Advances in Neural Information Processing Systems*, Dec 2025.

Partisan disparities in the funding of science in the United States. Alexander C Furnas, **Nic Fishman**, Leah Rosenstiel, and Dashun Wang. *Science*, Sep 18, 2025.

Estimating Vote Choice in U.S. Elections with Approximate Poisson-Binomial Logistic Regression. **Nic Fishman** and Evan Rosenman. *NeurIPS 2024 Workshop on Optimization for Machine Learning*, Dec 2024.

The fragility of fairness: Causal sensitivity analysis for fair machine learning. Jake Fawkes*, **Nic Fishman***, Mel Andrews, and Zachary Lipton. *Advances in Neural Information Processing Systems*, Dec 2024.

Human mobility networks reveal increased segregation in large cities. Hamed Nilforoshan*, Wenli Looi*, Emma Pierson*, Blanca Villanueva, **Nic Fishman**, Yiling Chen, John Sholar, Beth Redbird, David Grusky, and Jure Leskovec. *Nature*, Dec 21, 2023.

Metropolis sampling for constrained diffusion models. **Nic Fishman**, Leo Klarner, Emile Mathieu, Michael Hutchinson, and Valentin De Bortoli. *Advances in Neural Information Processing Systems*, Dec 2023.

Diffusion models for constrained domains. **Nic Fishman**, Leo Klarner, Valentin De Bortoli, Emile Mathieu, and Michael Hutchinson. *Transactions on Machine Learning Research*, Jul 2023.

Change we can believe in: Structural and content dynamics within belief networks. **Nic Fishman** and Nicholas T Davis. *American Journal of Political Science*, Jul 2022.

Should attention be all we need? The epistemic and ethical implications of unification in machine learning. Nic Fishman^{*} and Leif Hancox-Li^{*}. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, Jun 21, 2022.

Systematic characterization of generative models for de novo design of regulatory DNA. Nic Fishman, Avanti Shrikumar, Georgi Marinov, and Anshul Kundaje. *ICML Computational Biology Workshop (Spotlight)*, 2020.

Preprints and working papers

Distribution-Conditioned Transport. Nic Fishman^{*}, Gokul Gowri^{*}, Paolo LB Fischer^{*}, Marinka Zitnik, Omar Abudayyeh, and Jonathan Gootenberg. *arXiv preprint arXiv:2603.04736*, Mar 5, 2026.

Editorial Screening when Science is Cheap. Nic Fishman and Gabriel Sekeres. *Working paper*, 2026.

Making (Global) Criminal Procedure: Empire, the End of Justice, and the Rise of Efficiency. Nic Fishman. *Stanford University*, Aug 29, 2021.

Honors & Awards

- 2023 **James Mill Peirce Fellowship**, Harvard University
- 2023 **Graduate Research Fellowship**, National Science Foundation
- 2021 **Rhodes Scholarship**, Rhodes Trust
- 2021 **Outstanding Senior Thesis Award and Firestone Medal**, Stanford University

Teaching

STAT 221: Computational Statistics: Monte Carlo and Optimization Cambridge, MA
Section Leader, Harvard University; overall-item mean: 5.00/5 (8 responses) Spring 2026

STAT 236: Sparse Inference, and Network and Text Analysis Cambridge, MA
Section Leader, Harvard University; overall-item mean: 4.88/5 (8 responses) Fall 2024

Selected Talks

- 2025 **Generative modeling on the space of empirical measures**, Boston Diffusion Day; Harvard Statistics Seminar Cambridge, MA
- 2025 **False Promises and False Premises of Fair Machine Learning**, Columbia Algorithms and Society Seminar New York, NY
- 2025 **Predicting perturbations at single-cell resolution**, Cell Circuits and Epigenomics Seminar Cambridge, MA
- 2022 **Should attention be all we need?**, Stanford HAI; DeepMind Sociotechnical Systems Stanford / London
- 2022 **Making (Global) Criminal Procedure**, Law and Society Association Annual Meeting Lisbon, Portugal
- 2021 **Structural and content dynamics within belief networks**, American Political Science Association Annual Meeting Seattle, WA

Research Experience

Harvard Medical School, Harvard University Cambridge, MA
PhD research with Marinka Zitnik 2024–Present

Harvard Medical School, Harvard University Cambridge, MA
Research with Jonathan Gootenberg and Omar Abudayyeh 2024–Present

Institute for Quantitative Social Science, Harvard University Cambridge, MA
Research with Kosuke Imai 2020–2024

Center for Science of Science & Innovation, Northwestern University Evanston, IL
Research Fellow 2021–2023

Department of Economics, Massachusetts Institute of Technology Cambridge, MA
Research with Victor Chernozhukov 2021–2022

Professional Experience

Google Research Intern, Knowledge Graph Team	<i>San Francisco, CA</i> <i>Summer 2022</i>
Data for Progress Data Scientist	<i>New York, NY</i> <i>2018–2021</i>